From Probability Machine to Axiom Machine: Why Large Language Models Cannot Grasp the Rules of the World

—— From Symbol Fitting to Rule Anchoring

Guorui He
Independent Researcher

Corresponding author: Guorui He

Email: NooEcology@outlook.com
ORCID: 0009-0006-2947-0032
GitHub: https://github.com/wenhe1994/wenhe001

Abstract

C urrent large language models (LLMs) operate on statistical probability, fitting co-occurrence relationships among characters from massive text corpora, achieving remarkable results. However, their fundamental deficiency lies in the fact that they only learn “arrangements of symbols” without ever touching the “world rules behind the symbols.”

This paper first reviews mainstream approaches (scaling, retrieval-augmented generation, chain-of-thought, long-context, modular architectures, embodied AI) and argues that they all remain within the paradigm of “optimizing within an empirical space,” thus failing to fundamentally resolve issues such as hallucination, loss of coherence in long texts, and logical inconsistency. It then examines the nature of symbols, demonstrating that symbols are merely interfaces between human cognition and world rules; probabilistic fitting captures only superficial correlations between symbols and cannot establish deep structures such as causality, logic, or real-world constraints.

The paper further proposes a cognitive hierarchy leap: from symbols to concepts (compressing probability space to address the global workspace problem), and from concepts to axioms (logical constraints), showing how this leap fundamentally reshapes the capability boundaries of intelligent systems. Finally, it points to a fundamental solution: building a neuro-symbolic architecture anchored in an axiomatic system and a concept repository, thereby transforming models from “probabilistic prediction” to “rule-based reasoning.”

Keywords: Large language models; symbol fitting; concept anchoring; axiomatic system; neuro-symbolic fusion; intelligence architecture

1. Introduction

In 1846, the French astronomer Urbain Le Verrier, armed with nothing but pen and paper, deduced the position of Neptune from perturbations in the orbit of Uranus. He wrote to the Berlin Observatory; that same evening, telescopes turned to the coordinates he had calculated, and seventeen minutes later, Neptune was discovered [1]. Le Verrier never saw the planet with his own eyes, yet the mathematical rules he derived brought him closer to the truth than any direct observation could have.

The metaphor of this story reaches far deeper than its surface.

Today, we are attempting to “see” intelligence in a different way. We have built neural networks of unprecedented scale and fed them the largest text corpora in human history, allowing models to learn probabilities in an ocean of characters. They can write poetry, generate code, and hold conversations, as if they were approaching some form of “understanding.” Yet when we ask them to process long texts, perform multi-step reasoning, or answer questions that require deep causal inference, they unpredictably “go off the rails”—producing logically broken answers, fabricating facts that never existed, or even contradicting themselves within a single response. This is hallucination, loss of control—intractable ailments of current large language models (LLMs) [2,3].

Why can humans understand the world stably while models keep going astray? Mainstream technical approaches offer their own answers. Some believe the models are still not large enough, the data not abundant enough, and therefore continue down the path of scaling [4]. Others try to provide models with external knowledge bases so that they “look up information” before generating answers—a technique known as retrieval-augmented generation (RAG) [5]. Still others hope models can learn to “think step by step,” using chain-of-thought (CoT) to simulate reasoning [6], or employ long-context techniques to let them “see” further [7]. Some even argue that intelligence must be grounded through real-time interaction with the physical environment, leading to embodied AI [8].

Each of these approaches has achieved local progress, but they share a common feature: they all attempt to optimize probability itself, rather than reaching for rules. Scaling is a wider-range probabilistic fit; RAG expands the probability space with external texts; CoT generates stepwise within the probability space; long-context broadens the probabilistic horizon; embodied AI substitutes physical experience for symbol probability. All operate under a fundamental presupposition: that intelligence is a statistical model of symbolic associations.

This paper challenges that presupposition.

We will argue that symbols are merely interfaces between human cognition and the rules of the world, not the rules themselves. A symbol can be arbitrarily named (“apple” in Chinese, English, and Japanese are all different), yet the entity it denotes—the fruit with a growth cycle, edibility, and association with autumn—is governed by universal, stable rules. Current models learn only the co-occurrence probabilities among symbols; they never reach the rules to which symbols refer. This is like having a ruler that can measure length without knowing what length is—the model learns the graduations on the ruler, but not the quantity it measures.

This fundamental limitation leads to insurmountable difficulties for current technical paths: the probability space explodes as the semantic field expands, causing models to get lost in an infinity of combinations; causality is reduced to correlation, logic degraded to statistical trend; alignment can only be achieved through external “biasing” (e.g., reinforcement learning from human feedback, RLHF [9]), without embedding a stable reasoning framework.

This paper further argues that to escape the probabilistic labyrinth, we must accomplish two cognitive-level leaps: from symbols to concepts—compressing the probability space with concepts to address the attention explosion problem in the global workspace; and from concepts to axioms—using an axiomatic system to provide logical constraints on the relationships among concepts, thereby achieving a paradigm shift from “empirical memorization” to “rule-based inference.” We will use the simple question “In which season can apples in an orchard be picked and eaten?” as a case study to show how these two leaps transform a vague probabilistic problem into a clear logical deduction.

Finally, we will point out that the only way to fundamentally resolve the current predicament of AI is to build a neuro-symbolic architecture anchored in an axiomatic system and a dynamic shared symbolic concept repository. This is not a patch to existing technology; it is a re-foundation of the nature of intelligence—from “memorizing patterns” to “understanding rules.”

The structure of this paper is as follows: Section 2 reviews the nature of large language models, revealing their underlying logic as “character-probability machines.” Section 3 analyzes the nature of symbols, demonstrating that symbols are interfaces, not rules. Section 4 presents the first leap—from symbols to concepts—explaining how concepts compress probability space and solve the global workspace problem. Section 5 presents the second leap—from concepts to axioms—explaining how axioms provide logical constraints for reasoning. Section 6 critically examines mainstream technical approaches, showing why they have failed to achieve these two leaps. Section 7 summarizes the fundamental impasse. Section 8 outlines a way forward, sketching the axiomatic-driven neuro-symbolic architecture.

This work is grounded in the author’s System Ontology and Noetic Ecology theoretical systems [10,11], which provide an axiomatic foundation for understanding the hierarchical relations among symbols, concepts, and axioms. Due to space constraints, this paper focuses on critique of mainstream approaches and a directional blueprint; the complete architecture is elaborated in the monographs.

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2. The Nature of Large Language Models: Character-Probability Machines

To understand the fundamental limitations of current large language models, we must first recognize how they work. Despite their enormous scale and structural complexity, from the perspective of information processing they are essentially *character-probability machines*.

2.1 Predicting the Next Character

Modern large language models (such as the GPT series, Claude, Llama, etc.) are built on the Transformer architecture [1]. Their core task can be summarized as: given the preceding text, predict the probability distribution of the next character (or subword).

Let a text sequence be $w_1, w_2, \ldots, w_t$. What the model learns is the conditional probability $P(w_{t+1} \mid w_1, \ldots, w_t)$. During training, the model adjusts its internal parameters using massive text corpora to make this probability distribution approximate the distribution of real text as closely as possible. After training, the model possesses a fixed, enormous parameter matrix—which stores not “knowledge” but the statistical co-occurrence patterns of characters extracted from the data.

When a user enters a prompt, the model samples from this conditional probability distribution to generate the next character, then appends that character to the context and continues predicting the next, repeating this process until a complete response is produced.

This mechanism determines that all model behavior is based on statistical correlation, not logical necessity. For example, the model observes that “apple” is frequently followed by “eat,” “sweet,” “red,” and so on; thus it learns to establish probabilistic connections among these words. But it does not understand what “apple” is, nor does it understand what conditions “eat” requires—it only knows that, statistically, these words often appear together.

2.2 Attention Mechanism: Temporarily Delimiting the Semantic Field

One of the core innovations of the Transformer is the attention mechanism [2]. It allows the model, when predicting the next character, to “look back” at every preceding character and assign different weights to them according to their relevance to the current prediction.

The function of this mechanism can be understood through a simple metaphor: the attention mechanism temporarily delimits a *semantic field*. Within this field, the original weights of each character are either activated or suppressed, enabling the model to “focus” on information most relevant to the current context while ignoring irrelevant noise.

For instance, when a user enters “In which season can apples in an orchard be picked and eaten?”, the attention mechanism causes the model, when predicting “season,” to pay more attention to words such as “orchard,” “apple,” “pick,” and “eat,” while relatively ignoring function words like “in,” “can,” etc. This amounts to a probabilistic collapse of directions in the character-probability space, making “autumn” more likely and “spring” less so.

However, this collapse is still probabilistic. It merely adjusts weights temporarily based on statistical features of the preceding text; it does not truly understand the entities to which the characters refer or the rules behind them. Even if the attention mechanism links “apple” to “autumn,” the model does not know *why* apples are harvestable in autumn—it only knows that this association appears frequently in the corpus.

2.3 The Illusion of “Understanding”

Large language models create the illusion of “understanding” because, in most everyday scenarios, the statistical patterns of language closely coincide with the real-world patterns. When people say “apple,” it is often followed by “eat,” so the model learns “apple → eat.” When such associations appear frequently in the training data, the model can produce text that aligns with common sense.

But the problem is that statistical correlation is not causal necessity. The model can learn the statistical association between “apple” and “autumn,” but it cannot distinguish whether this association is causal (apples are harvestable in autumn because they ripen then) or merely coincidental (the two words happen to co-occur in some texts). When faced with scenarios requiring causal reasoning, the model reveals its true nature: it is only “memorizing patterns,” not “understanding them.”

Worse still, when the semantic field expands, the probability distribution of characters becomes extremely complex, and the attention mechanism cannot maintain global coherence. At each step the model selects the locally optimal next character, but the accumulation of local optima does not guarantee global logical correctness. This is precisely the root cause of long-text incoherence, logical breakdowns, and hallucinations.

2.4 The Gap Between “Memorizing” and “Understanding”

We can illustrate the gap between current models and genuine intelligence with an analogy:

- *Memorizing the multiplication table*: A person can recite “three times three is nine, four times four is sixteen,” yet may not understand that multiplication is a convenient form of addition, let alone the axioms behind commutativity or associativity.
- *Understanding multiplication*: A person comprehends the definition of multiplication, its properties, and can derive new results for unseen combinations, even prove the commutative law.

Current large language models are like the person who has memorized the multiplication table. They have learned countless “linguistic formulas” from vast text corpora and can respond fluently in common scenarios. But once they encounter situations that require deduction, causal chains, or logical consistency, they reveal themselves to be merely “memorizing” rather than “understanding.”

This is precisely the fundamental predicament that the rest of this paper will address: fitting the co-occurrence probabilities of symbols alone can never reach the world rules behind the symbols. To cross this chasm, we must re-examine the nature of symbols and construct a cognitive architecture capable of carrying rule-based reasoning.

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3. The Nature of Symbols: Interfaces, Not Rules

The fundamental limitation of large language models stems from a deep misunderstanding of the nature of symbols. To see this limitation clearly, we must revisit the basic insights of semiotics and push further: what exactly is the relationship between symbols and the world?

3.1 The Arbitrariness and Conventionality of Symbols

Ferdinand de Saussure, the founder of modern semiotics, pointed out that a symbol is composed of a *signifier* (the sound or written form) and a *signified* (the concept). The bond between them is *arbitrary* [1]. Different languages use entirely different sounds to refer to the same thing—"apple" is "píngguǒ" in Chinese, "apple" in English, "りんご" in Japanese. There is no natural, necessary connection between signifier and signified; their correspondence is a product of social convention.
This insight reveals a crucial fact: symbol systems themselves do not carry any necessary truth. Symbols can be arbitrarily replaced; as long as a community agrees, the new symbol will work. Yet once a symbol system is established, it imposes constraints—individuals cannot change it arbitrarily, otherwise they will not be understood. The unity of arbitrariness and constraint constitutes the basic characteristic of symbols.

3.2 Wittgenstein's Insight and Its Limit

Ludwig Wittgenstein pushed this line of thinking further in his *Philosophical Investigations*. He introduced the concept of "language games," arguing that the meaning of a word lies in its *use*, rather than in a correspondence with some object [2]. The meaning of a word is the rule for its use in language.
Wittgenstein's contribution was to free symbols from the simple function of "naming objects," revealing the dynamism, context-dependence, and rule-governed nature of meaning. However, his theory stopped at this point—he described how language works, but did not answer a deeper question: *why* does language work? *Why* can symbols refer to the world?

3.3 Symbols as Interfaces: The Ruler Analogy

To answer this question, we must place symbols within a broader ontological framework.
This paper proposes that **symbols are interfaces between human cognition and the rules of the world**.[1]
Just as a ruler can measure length but is not itself length, a symbol can denote things in the world but is not itself those things. The graduations on a ruler are arbitrary (they can be in centimeters, inches, or even a custom unit), yet the length they measure is objective. Similarly, the form of a symbol is arbitrary (different languages use different symbols), but the world rules it points to are universal.
This "interface" perspective contains three layers:

- **World rules**: objective regularities, such as causality, physical laws, biological cycles. These rules exist independently of any symbol.
- **Symbol systems**: conventional tools created by humans to describe world rules. They can change, evolve, and even differ radically across cultures.
- **Cognitive mediation**: humans understand world rules through symbol systems; symbols are both the bridge to understanding and the boundary of understanding.

An apple can be named "apple," "pomme," or "苹果" in different languages, but the apple's growth cycle, its edibility, and its association with autumn—these rules do not change when the symbol changes. Symbols are only channels through which we approach these rules; they are not the rules themselves.

[1] For a systematic treatment of symbols as cognitive interfaces, see the author's System Ontology (He, 2026a), particularly the axioms of "Material/Noetic Duality" and "Symbols as Noetic Genes."

3.4 Where Do Symbol Weights Come From?

Current large language models, trained on massive text corpora, learn the co-occurrence probabilities among symbols. The probability that a character "should" appear in a given context is encoded into the model's parameters. Yet these "weights" are not inherent properties of the symbols themselves; they are statistical traces of how humans have used them.

When a community consistently uses "apple" to denote the red, edible fruit and talks about its ripening in "autumn," a statistical association forms between "apple" and "autumn." What the model fits is precisely this association. But what the model does not know is that this association exists because there is an objective causal chain behind it: apple trees bear fruit in autumn, and when the fruit ripens it can be picked and eaten.

The weights of symbols are traces left by long-term human usage, not attributes of the symbols themselves. The model learns the traces but never touches the source of the traces. To overcome this limitation, we must anchor symbols to the concepts and rules behind them—the very cognitive leap from symbols to concepts, and from concepts to axioms, that this paper will develop.

In the author's theory of Noetic Ecology, this process is formalized as the three-tier model of consciousness and the theorem of cognitive escape (He, 2026b).

3.5 Core Conclusion

We can now state the core conclusion of this section:

Symbols themselves carry no rules. Rules are in the world, not in the symbols.

Symbols are interfaces between human cognition and the rules of the world. Their form is arbitrary, but their validity depends on whether they accurately point to the rules. Fitting only the co-occurrence probabilities of symbols captures the statistical traces of human symbol use but can never reach the world rules behind the symbols.

This means that large language models based on probabilistic fitting, no matter how large they grow or how abundant their data become, will never truly "understand" the world. They can learn the statistical association "apple → autumn," but they do not know why apples ripen in autumn; they can produce sentences like "Because apples ripen in autumn, they can be picked," but they do not grasp the causal meaning carried by "because … therefore … ."

This limitation is not a missing technical detail; it is a fundamental paradigm flaw. To escape the probabilistic labyrinth, we must accomplish the cognitive leap from *symbol fitting* to *rule* *anchoring*. This is the task to which the following sections will turn.

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4. From Symbols to Concepts: The First Cognitive Leap — Probability Compression and the Global Workspace

The previous section argued that current large language models can only fit co-occurrence probabilities among symbols and cannot reach the world rules behind them. Yet even if we acknowledge this limitation, a deeper question remains: why do humans, when faced with complex contexts, not fall into the same probabilistic explosion as large models? Humans also need to process vast amounts of information; how do they maintain logical consistency? The answer lies in a crucial intermediate layer of human cognition—*concepts*.

4.1 The Probabilistic Explosion Problem: Why Attention Is Not Enough

The core computational unit of large language models is the attention mechanism. At each step, it attempts to sift through all preceding characters to select relevant information, assign weights to each, and then predict the next character based on the weighted information. When the context is short (e.g., tens of characters), attention works effectively. But as text length grows, problems become acute:

- **Information dilution**: As the number of characters increases, the average weight per character declines. Truly critical information may be drowned out by noise.
- **Combinatorial explosion**: The number of possible next-character combinations grows exponentially. At each step the model selects a local optimum, but the accumulation of local optima does not guarantee global logical coherence.
- **Long-range dependency failure**: When causal chains span hundreds of characters, the attention mechanism struggles to maintain stable associations, leading to logical breaks.

This is the root cause of uncontrollable long-text generation and frequent hallucinations. The attention mechanism can only handle “local correlations”; it cannot maintain “logical necessity” across a global space. Attempts to alleviate this problem by expanding context windows or improving attention computation are essentially more refined searches *within* the probability space, but they never escape the cage of “probability.”

4.2 The Compressive Nature of Concepts

The human cognitive system solves this problem by introducing an intermediate layer—*concepts*. A concept is a compression of the probability space; it collapses the infinite combinatorial paths of character-level possibilities into a finite network of relations.

Take “apple” as an example. At the character level, the possible combinations involving “apple” are nearly infinite: colors, sizes, origins, varieties, ways to eat, cultural associations … any one direction is diluted within an enormous space. But once we recognize the *concept* “apple,” the situation changes completely:

- The concept “apple” anchors a stable set of properties: it is a fruit, edible, has a growth cycle, is associated with autumn, can be picked.
- These properties are not random probability distributions; they are necessary connections determined by world rules. Apples ripen in autumn because of their biological characteristics, not because “apple” and “autumn” co-occur frequently in corpora.
- Relationships among concepts are sparse and structured. A concept has only a few key relationships (e.g., “apple → autumn,” “apple → edible”), far fewer than the infinite combinatorial possibilities at the character level.

Thus, a concept is essentially a compression of the probability space. It collapses the infinite possibilities of the character layer into a finite set of concept nodes and the structured relations among them. Once a problem is elevated to the conceptual layer, the system shifts from “drifting in a probabilistic ocean” to “navigating in a relational network.”

4.3 How Concepts Solve the Global Workspace Problem

The concept of “working memory” from cognitive psychology provides a useful perspective [1]. Working memory is the “mental space” where we process information in real time; its capacity is limited (typically handling about seven chunks at once). If we were to process at the character level, working memory would soon be overwhelmed by a flood of details. But through conceptual compression, we can package large amounts of character-level information into a single concept, thereby accommodating more content in the working memory space.

For example, to understand the sentence “In which season can apples in an orchard be picked and eaten?” the human cognitive system does not analyze the statistical probability of each character one by one. Instead, it quickly identifies a handful of concepts: *orchard*, *apple*, *pick*, *eat*, *season*. These concepts themselves already carry extensive background knowledge: an orchard is a place where fruit is grown; apples ripen in autumn; picking apples requires ripeness … This knowledge is not learned on the fly from the current context; it is embedded in the concepts themselves. Thus, the global workspace need only handle these few concepts and their relationships, avoiding the probabilistic explosion of the character layer.

Large language models lack this capability. They have no built-in concept repository; all information is stored as probability distributions over characters. When they must handle long texts, they can only search probabilistically at the character layer, inevitably falling into combinatorial explosion.

4.4 From Symbol to Concept: The Necessity of Anchoring

Concepts do not arise spontaneously. For a symbol to become a concept, it must be *anchored*—that is, a stable mapping must be established between the symbol and the world rules behind it.

Consider “apple.” At the symbol level, “apple” is just a sequence of five characters that could refer to anything. At the concept level, “apple” must be bound to the entity that has a specific growth cycle, is edible, and is associated with autumn. This binding is not a statistical correlation; it is a rule-based linkage, grounded in human understanding of the world: apple trees blossom in spring, bear fruit in summer, and ripen in autumn. These regularities are objective and stable; they do not change when the language changes.

Current large language models have not accomplished this anchoring. They know that “apple” and “autumn” frequently co-occur, but they do not know why. When asked “In which season can apples in an orchard be picked and eaten?” they might output “autumn,” but they might also output “spring” or “winter” if the corpus contains a few unusual statements. Even if the answer is correct, it is merely a successful probabilistic sample, not a rule-based deduction.

4.5 Case Study: Apples in an Orchard

Let us use a concrete example to illustrate how the leap from symbols to concepts transforms the reasoning process.

Character-level processing (current large models):

Input: “In which season can apples in an orchard be picked and eaten?”

The model encodes each character as a vector, uses the attention mechanism to compute the strength of association between “pick” and “apple,” then between “season” and “pick,” and so on. At each step it chooses the most probable path in an enormous probability space. Because “apple” and “autumn” co-occur frequently in the corpus, the model may output “autumn.” But this result is not necessary: if the corpus contains statements like “apples blossom in spring,” the model might mistakenly associate “spring” with “pick.”

Concept-level processing (the direction proposed in this paper):

The system first identifies the key concepts: orchard, apple, pick, eat, season. These concepts are anchored to nodes in a Dynamic Shared Symbolic Concept Bank (DSSCB). Each node carries its definition and relational edges:

- Apple: fruit, edible, growth cycle: blossoms in spring → bears fruit in summer → ripens in autumn.
- Pick: action, precondition: fruit must be ripe.
- Season: temporal concept, linked to biological cycles.

The system does not search at the character level; instead, it searches for paths in the concept graph: apple → growth cycle → autumn → ripe → pickable → edible. The reasoning result is deterministic: only autumn satisfies the causal chain. This result does not depend on statistical frequencies in the corpus; it is guaranteed by the rules embedded in the concepts themselves.

4.6 The Cost and Value of Conceptual Compression

Elevating symbols to concepts is not cost-free. It requires building a concept repository that covers a broad range of domains, that can evolve over time, and establishing stable mappings between symbols and concepts. This demands a substantial knowledge engineering effort that cannot be achieved overnight.

Nevertheless, the cost is worth paying. Once the leap from symbols to concepts is accomplished, the system gains several fundamental advantages:

- **Logical consistency**: Relationships among concepts are governed by rules, not statistical correlations, ensuring stable reasoning.
- **Global controllability**: The amount of information at the conceptual level is far smaller than at the character level, making reasoning within a global workspace feasible.
- **Explainability**: Reasoning can be traced back to paths in the concept graph; every step is justifiable.
- **Evolvability**: The concept repository can be updated dynamically; new knowledge can be incorporated into the existing relational network without retraining the entire model.

This marks the first step in the paradigm shift from "probabilistic prediction" to "rule-based reasoning."

4.7 Toward the Next Leap: From Concepts to Axioms

Conceptual compression solves the global workspace problem, but concepts themselves still have limitations. Relationships among concepts remain *factual*—they describe "what is," not "why." For example, a concept graph might contain both "apple → autumn" and "apple → all seasons" as edges, and the system would have no way to decide which is correct. To overcome this, we need a higher-level constraint—an *axiomatic system*.

An axiomatic system provides logical rules (such as causality, non-contradiction, the law of excluded middle) that govern relationships among concepts. It elevates factual knowledge into logical necessity, transforming reasoning from empirical memory to rule-based inference. The next section will elaborate this second leap.

References

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5. From Concepts to Axioms: The Second Cognitive Leap — Logical Constraints

Conceptual compression solves the global workspace problem, enabling the system to process complex information within the limited capacity of working memory. Yet concepts themselves have a fundamental limitation: the relationships among concepts are factual rather than logical. A concept graph may contain both "apple → autumn" and "apple → all seasons" as edges, but it cannot determine which is correct. To complete the transition from "empirical memory" to "rule-based inference," a higher-level constraint must be introduced—an *axiomatic system*.

5.1 The Limitation of Concepts: The Predicament of Factual Knowledge

Concepts and their relational networks are, by nature, organizations of factual knowledge. They tell the system "what is what": an apple is a fruit; autumn is a season; apples ripen in autumn. But these statements are static and descriptive; they cannot answer the more fundamental question: *why* do apples ripen in autumn? Why is one relationship more reliable than another?

When contradictions exist in the concept graph (e.g., "apple → autumn" coexisting with "apple → all seasons"), factual knowledge alone provides no basis for choice. The system would have to fall back on statistical frequency—choosing the relationship that appears more often—which returns it to the probabilistic cage.

Worse, factual relationships cannot handle causality. Causality requires a necessary "cause → effect" linkage, whereas factual relationships can only give correlations. Apples correlate with autumn, but correlation does not imply causation: does autumn cause apples to ripen, or does ripening cause autumn? Without causal constraints, a concept graph can only describe co-occurrences; it cannot infer necessity.

5.2 The Nature of an Axiomatic System: Constraints at the Level of Rules

An axiomatic system constitutes the second cognitive leap. It elevates factual knowledge into logical necessity, providing hard constraints on the relationships among concepts.

An axiomatic system consists of a set of self-evident fundamental propositions that define the legitimate rules of inference. In traditional logic, axioms include the law of identity (*A* is *A*), the law of non-contradiction (*A* and not-*A* cannot both be true), and the law of excluded middle (*A* or not-*A* must be true). In a broader sense, an axiomatic system can encompass domain-specific foundational principles, such as causality, conservation laws, and moral principles.

The relationship between the axiomatic system and the concept network can be summarized as follows: concepts are nodes of knowledge; axioms are the inference rules that govern the relations among those nodes. The concept network tells the system what things exist and what properties they have; the axiomatic system tells the system how to derive new knowledge from those things and their properties, and how to reject logically illegitimate inferences.

5.3 How Axioms Prune Illegitimate Paths

Take again the question "In which season can apples in an orchard be picked and eaten?" At the conceptual level, the system identifies the concepts "apple," "autumn," "pick," "eat," and finds the relation "apple → autumn" in the concept graph. But there may be other paths in the graph, such as "apple → imported → available all year." How can the system rule out this path?

The axiomatic system plays the decisive role here. Suppose the axiomatic system includes the following rules:

- **Axiom of causal consistency**: any valid inference must be based on a verifiable causal chain. Effects cannot precede causes, and there can be no effect without a cause.
- **Axiom of contextual matching**: inference must be consistent with the current task context; reasoning detached from context is invalid.

In the path "apple → imported → available all year," the causal chain requires that "available all year" is caused by "imported." But the current context is "apples *in an orchard*," not "imported apples." The axiom of contextual matching deems this path incompatible with the current context and thus excludes it. In contrast, the path "apple → autumn → ripe → pickable → edible" has a complete causal chain (autumn causes ripeness, ripeness makes picking possible), and the context matches (apples in an orchard naturally follow a growth cycle). Hence it is adopted.

Without an axiomatic system, the concept graph can only present all possible relationships; it cannot determine which relationships are legitimate in the current context. The axiomatic system provides the criteria for such judgment.

5.4 From Facts to Rules: The Cognitive Leap

Introducing an axiomatic system marks the transition from "factual memory" to "rule-based inference":

- *Factual memory*: remembering that "apples are pickable in autumn." This is the level of empirical capability, which large language models already possess.
- *Rule-based inference*: understanding that "apples are pickable" because "apples ripen," and "apples ripen" because "autumn is part of the apple's growth cycle." This is the level of logical capability, requiring an axiomatic system to guarantee the legitimacy of the

inferential chain.

The fundamental value of this leap lies in the fact that once the rules are grasped, the system can handle novel situations it has never encountered. It does not need to rely on whether the corpus contains the specific statement "apples in an orchard are pickable in autumn"; it can deduce "autumn is the harvest season" from the rule "apples have a growth cycle." When asked "In which season can lychees be picked?" it can similarly reason based on analogous rules (lychees ripen in summer), even if the training corpus never contained the specific phrase "lychees are pickable." This is the hallmark of rational intelligence: deriving an infinity of possibilities from a finite set of rules.

5.5 The Division of Labor: Concepts and Axioms

The concept network and the axiomatic system each have distinct roles, complementing each other:

- The **concept network** stores factual knowledge: an apple is a fruit, autumn is a season, an orchard grows apples. This knowledge can evolve, be updated, and even differ across cultures.
- The **axiomatic system** defines logical rules: causality, non-contradiction, contextual matching. These rules are stable and domain-independent, providing the underlying framework for the use of factual knowledge.

Concepts can evolve; axioms must remain stable. The concept of an apple can be updated with advances in agricultural technology (e.g., new varieties with different ripening seasons), but causality cannot be overturned because an experiment fails. It is precisely the stability of the axiomatic system that prevents the dynamic evolution of the concept network from causing cognitive collapse.

5.6 From Concepts to Axioms: The Complete Cognitive Hierarchy

We can now outline the complete cognitive hierarchy of intelligence:

- **Symbol layer**: the arrangement and combination of characters, surface associations driven by probability. This is the level at which current large language models operate.
- **Concept layer**: symbols anchored to stable concept nodes, forming a relational network among concepts. This solves the global workspace problem, enabling the system to handle complex information within limited capacity.
- **Axiom layer**: logical rules constraining the relationships among concepts, achieving the transition from factual memory to rule-based inference. This is the core of rational intelligence.

These three layers are not independent. The concept layer compresses the information from the symbol layer into operational knowledge units; the axiom layer provides the criteria of legitimacy for reasoning at the concept layer. Together they constitute a complete cognitive system capable of extracting rules from symbolic input and inferring new knowledge under the constraints of those rules.

5.7 Toward the Next Section: Critique of Mainstream Approaches

This section has argued that the fundamental limitation of current large language models, and the two cognitive leaps required to overcome it, point to a clear direction. The next task is to apply this analytical framework to mainstream technical approaches, examining each in turn to show why they have failed to achieve these leaps and therefore cannot resolve the fundamental problem of probabilistic fitting. Section 6 will undertake this critique.

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6. Critique of Mainstream Approaches: Why They Remain Trapped in the Labyrinth

Faced with the fundamental limitations of large language models, researchers have proposed a variety of improvement paths. While these approaches have achieved considerable engineering progress, a closer examination reveals that each remains within the paradigm of "optimizing within a probability space." None has achieved the cognitive leaps from symbols to concepts and from concepts to axioms. This section will critically examine these mainstream approaches and show why they cannot fundamentally resolve problems such as hallucinations, long-text incoherence, and logical inconsistency.

6.1 Scaling: Larger-Scale Probabilistic Fitting

Scaling is the most intuitive approach: increase model parameters, expand training data, add more compute, hoping that "brute force" will make intelligence emerge [1]. In the short term, this approach has indeed brought performance gains. Yet its fundamental assumption—that larger-scale probabilistic fitting can approximate genuine understanding—has been disproven both theoretically and empirically.

Theoretically, scaling merely expands the search range within the probability space; it does not change the nature of the model. The model still fits co-occurrence probabilities among characters, still lacks concept anchoring, and still has no axiomatic constraints. When the corpus contains enough instances of "apples ripen in autumn," the model can output the correct answer with high probability. But once it encounters causal chains that are rare in the corpus, or must rule out erroneous paths through multi-step reasoning, the model still falls into probabilistic explosion.

Empirically, the marginal returns of scaling are diminishing [2]. Hallucinations and logical errors have not disappeared as model size increased; instead, they have become more prominent in longer texts and more complex reasoning tasks. Scaling only enlarges the labyrinth; it does not install an exit.

6.2 Retrieval-Augmented Generation: Expanding the Probability Space with External Text

Retrieval-augmented generation (RAG) attempts to alleviate hallucinations by introducing external knowledge bases [3]. Before generating an answer, the model retrieves relevant documents from a knowledge base and uses them as part of the context for generation. This can indeed reduce factual errors, but it does not touch the root problem.

The external knowledge in RAG is still *text chunks*, not anchored concepts. The model still processes the retrieved text probabilistically, treating it as ordinary characters within the probability calculation. It cannot distinguish which information is factual and which is logical, let alone integrate the retrieved knowledge into stable conceptual relations. When the retrieved text conflicts with the model's existing probability distribution, the model may still choose the more statistically common erroneous path.

More fundamentally, RAG cannot handle open-ended questions that require reasoning. Even if it retrieves the fact that "apples ripen in autumn," the model still does not understand the causal chain; when the question becomes "Would the harvest season be the same for apples grown in a greenhouse?" RAG is powerless—the retrieval base may contain no direct answer, and the model lacks the capacity to reason from principles.

6.3 Chain-of-Thought and Reasoning Chains: Stepwise Generation within the Probability Space

Chain-of-thought (CoT) and its variants attempt to simulate human reasoning by having the model output intermediate reasoning steps before the final answer [4]. These methods have achieved notable success on tasks such as mathematics and logic, but their essence remains within the probability framework.

Every step of CoT is still probabilistic character prediction. When the model outputs "Because apples ripen in autumn, they can be picked in autumn," the generation of this sequence is still based on statistical associations, not logical necessity. The model may output correct reasoning steps, but it may also output erroneous steps that happen to be statistically common. As the reasoning chain grows longer, the error probability accumulates at each step, ultimately leading to overall failure—precisely why CoT breaks down in long-chain reasoning.

A deeper issue is that CoT has no built-in logical rules to constrain the reasoning process. The model may output a logically broken sentence like "Because apples ripen in autumn, they can be picked in spring," as long as such a sequence occasionally appears in the training corpus. Probabilistic fitting cannot distinguish correct reasoning from incorrect reasoning; it only reflects statistical frequencies.

6.4 Long Context and Sparse Attention: Widening the Probabilistic Horizon

Long-context techniques attempt to extend the attention window, allowing models to "see" further in order to maintain long-range dependencies [5]. Sparse attention optimizes computation, making long contexts feasible. While these techniques are significant engineering achievements, they still fail to escape the probability framework.

As the semantic field expands, the probability distribution of characters becomes increasingly complex. The attention mechanism can only focus on local correlations; it cannot guarantee global logical coherence. Even if the model can "see" all preceding characters, it still does not understand the entities those characters refer to or the rules governing them. Long contexts merely enlarge the space for probabilistic search without providing direction for that search.

6.5 Modular and Neuro-Symbolic Approaches: Shallow Integration

Some research attempts to combine neural networks with symbolic systems—for example, enhancing reasoning with external knowledge graphs [6], or embedding symbolic rules into neural networks [7]. These approaches have touched on the direction of neuro-symbolic fusion, but most

remain at the level of shallow integration.
A typical pattern is to let the neural network generate symbols (e.g., entities, relations) and then pass them to a symbolic system for processing. This "neural-first, symbolic-later" serial architecture cannot achieve real-time interaction between the neural and symbolic sides. The results of symbolic reasoning cannot feed back to the neural network, so subsequent predictions may still deviate from the correct path. More importantly, these attempts usually lack a unified semantic space; the concepts in the symbolic system and the vector space of the neural network are not directly commensurable, leading to information loss in transmission.
The architecture proposed in this paper differs fundamentally from these attempts: it builds a shared semantic space (DSSCB) that enables bidirectional real-time interaction, and introduces an axiomatic system as the underlying logical constraint, making symbolic reasoning not a mere add-on but the core decision-making mechanism.

6.6 Embodied AI: Acquiring Instincts through Physical Interaction

Embodied AI argues that intelligence must be built through real-time interaction with the physical environment; meaning must be "grounded" in action [8]. This approach has made progress in robotics and simple physical tasks, but when viewed as a general solution to intelligence, it suffers from fundamental limitations.
Through massive interactive experience, embodied AI solidifies "sensation-action" patterns into memories akin to conditioned reflexes. This amounts to memorizing the laws of the physical world *as experiences*, rather than abstracting them into inferable axioms. A trained robot dog can learn to avoid obstacles, but it does not understand the principle "avoid obstacles," nor can it apply it to unseen situations (e.g., reasoning in a virtual environment, understanding metaphors).
The intelligence produced by embodied AI is *instinctual*. It imitates the sensorimotor knowledge animals acquire through their bodies, not the rational knowledge humans acquire through language, logic, and mathematics. It cannot handle abstract symbols, perform causal inference, or maintain logical consistency—precisely what current large language models lack and what rational intelligence requires.
Embodied AI and symbol-probability machines may appear to follow different paths, yet they share the same fundamental defect: both merely *memorize patterns*. The former memorizes sensorimotor patterns, the latter memorizes statistical patterns of symbols. Neither has built a rule system capable of inferring the unknown.

6.7 Summary: The Common Predicament—No Framework

Looking across all the approaches surveyed above—whether scaling, RAG, CoT, long context, shallow neuro-symbolic fusion, or embodied AI—none have departed from the paradigm of "optimizing within an empirical space." They either perform a broader search in probability space, expand the probability space with external texts, simulate reasoning by stepwise generation, enlarge the probabilistic horizon, or substitute physical experience for symbol probabilities. Their common feature is that they have not installed a stable, unshakeable *framework* in the intelligent agent.
Current large language models have no framework; they have only a probability field. Embodied agents have no framework; they have only an experience base. Without a framework, causal consistency cannot be guaranteed; without a framework, logical contradictions cannot be ruled

out; without a framework, stable navigation in an infinite space of possibilities is impossible. This is the fundamental reason why all mainstream paths remain trapped in the labyrinth.

The next section will summarize this fundamental impasse and point to the only way out: installing an axiomatic system, establishing concept anchoring, and achieving deep integration between neural and symbolic processing.

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7. The Fundamental Impasse: Why Neither Probabilistic Fitting nor Embodied Experience Can Reach the Rules

The previous section examined each mainstream technical approach and showed that, despite their respective strengths, all remain within the paradigm of "optimizing within an empirical space." This section goes a step further, asking from a foundational perspective: why can none of these approaches reach the world's rules? Why can they only "memorize patterns" and never "understand the rules"?

7.1 The Nature of Probabilistic Fitting: Memorizing Statistical Patterns of Symbols

Large language models learn from massive text corpora the co-occurrence probabilities of symbols. They count how often "apple" is followed by "eat," how often "autumn" co-occurs with "ripe," and encode these statistical associations into their parameters. When a user asks a question, the model generates a character sequence along the highest-probability path.

This process is structurally isomorphic to a child memorizing the multiplication table. A child who has memorized "three times three is nine" can answer quickly, yet may not understand that multiplication is a convenient form of addition, let alone the axioms behind commutativity or associativity. He is simply "memorizing" a fixed set of answers.

Large language models are no different. They have memorized countless “linguistic formulas”: apples are edible, autumn brings ripeness, causal reasoning uses “because … therefore …” When a question falls within the memorized patterns, they can respond fluently; once they go beyond the boundary and need to infer new situations or new combinations, they reveal that they can only memorize, not think.

The deeper problem is that the statistical patterns of symbols are only *shadows* of the world’s rules, not the rules themselves. Apples ripen in autumn not because “apple” and “autumn” frequently appear together in corpora, but because of the biological characteristics of apple trees. Statistical patterns can describe the phenomenon, but they cannot explain it; they can simulate it, but they cannot infer from it.

7.2 The Nature of Embodied Experience: Memorizing Sensorimotor Patterns

Embodied AI attempts to ground meaning through physical interaction with the environment. A robot tries repeatedly, learns that “touching fire hurts” and “bumping into a wall is painful,” and hardens these experiences into behavioral patterns. This is analogous to how animals learn survival skills through conditioning.

Yet embodied experience is still *memorizing patterns*—it memorizes associations between sensations and actions. An animal that has learned to avoid fire knows “fire → pain → avoid,” but it does not know that fire is an oxidation reaction, that pain is a neural signal, or the causal chain behind the behavior “avoid.” It has merely memorized the association between “fire” and “pain,” just as the large language model memorized the association between “apple” and “autumn.”

The limitation of embodied AI is especially evident at the symbolic level. It cannot handle the metaphorical use of “fire” (e.g., “his anger flared”), cannot grasp abstract relations like causality, and cannot perform logical inference within a symbolic system. What it acquires is *instinctual intelligence*—a capacity isomorphic to animal behavior, not the rational intelligence unique to humans.

7.3 Common Defect: No Framework

Whether it is a symbol-probability machine or an embodied agent, the common defect is the absence of a *framework*.

Here “framework” refers to two layers of stable structure:

- **Conceptual framework**: anchoring symbols to stable concept nodes, so that symbols are no longer floating signifiers but point to entities with determinate properties and relations.
- **Axiomatic framework**: using logical rules to constrain relationships among concepts, so that reasoning is no longer probabilistic search but rule-based inference.

Without a conceptual framework, symbols remain floating signifiers, capable of being arbitrarily combined into “a blue sun at night.” Without an axiomatic framework, causality is reduced to correlation, logic to statistical trend.

Large language models have a probability field but no concept anchoring; embodied agents have an experience base but no axiomatic constraints. Both are *memorizing*—the former memorizing statistical patterns of symbols, the latter memorizing causal patterns of sensation and action. No matter how much they memorize, they cannot leap to a rule system that can infer the unknown.

7.4 Why Rules Can Only Be Captured through Axioms

The rules of the world are objective, stable, and inferable. Apples ripen in autumn not because humans write it that way in corpora, but because biological laws determine it. Water boils at 100°C not because humans describe it in texts, but because physical laws determine it.

Such rules can only be grasped through an *axiomatic system*. Axioms are self-evident fundamental propositions that define the legitimate rules of inference. Starting from axioms, we can derive an infinity of concrete facts without having to memorize each one individually. This is the essence of rational intelligence: deriving infinite possibilities from finite axioms.

Current large language models and embodied agents are following the opposite path: they attempt to approximate the finite rules of the world with infinite empirical memory. Theoretically, this path cannot succeed—the world's rules are infinite in their applications, yet any finite collection of experiences can never exhaust all possible applications of those rules.

7.5 Redefining the Alignment Problem: Having No Framework Is Itself a Framework

With this analysis, we can redefine the "alignment problem." The mainstream view holds that alignment is making an AI's goals consistent with human values [1]. But this view presupposes that the AI already has some kind of "goal framework." The analysis in this paper shows that this presupposition is false: current AIs have no framework at all. How then can we speak of "alignment"?

Yet a deeper insight emerges: **having no framework is itself a framework**.

When a system has no built-in axioms and no stable concept anchoring, its behavior is entirely determined by the statistical inertia of its training data. This statistical inertia itself constitutes a hidden, unspoken "framework"—it determines how the system responds to inputs, how it judges logic, how it implicitly ranks values. But this framework is invisible, uncontrollable, and impossible to correct from within.

The author has predicted in earlier work that this "framework-less" state will inevitably make alignment problems worse, and they will appear in increasingly subtle forms [2]. As models scale up and capabilities increase, they will behave in ways that are statistically consistent over a wide range, appearing superficially "aligned," yet this is merely statistical inertia masquerading as values. When encountering situations outside the statistical distribution, this pretense collapses, revealing that the system has no intrinsic value anchor. This is why the alignment problem is not a technical oversight but an inherent, unavoidable predicament of the current paradigm.

The true alignment problem should be: how to give AI a foundational framework that aligns with the rules of the world. This framework must include:

- A stable **axiomatic system** as the underlying logic for reasoning;
- An **evolving concept network** as the organizational structure of knowledge;
- **Deep integration of neural and symbolic processing**, so that probabilistic perception can be transformed into rule-based inference.

This is not a patch to existing models; it is a reconstruction of the nature of intelligence. It is not about making AI more "human-like," but about giving AI a cognitive architecture isomorphic to human rationality.

7.6 Toward a Way Out: From Memorizing Patterns to Understanding Rules

The conclusion of this section is sobering, yet clear: none of the current mainstream approaches can reach the world's rules; therefore none can achieve true rational intelligence. However, this conclusion also points to the only way forward: shift from memorizing patterns to understanding

rules, from probabilistic fitting to rule-based inference.
This is not an incremental improvement but a fundamental paradigm shift. It requires abandoning the habitual thinking of "optimizing within an empirical space" and returning to the first principles of intelligence: what is the nature of intelligence? What is the relation between symbols and the world? How can rules be grasped?
The next section will outline a new technical path based on the analysis in this paper—an axiomatic-driven neuro-symbolic architecture. This path does not reject existing technologies; it transcends their fundamental limitations.

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8. The Way Out: An Axiomatic-Driven Neuro-Symbolic Architecture

The analysis above reveals a fundamental predicament: neither symbol-probability machines nor embodied agents can reach the world's rules; they remain stuck at the level of "memorizing patterns." To escape this predicament, we must undergo a true paradigm shift—from probabilistic fitting to rule anchoring, from empirical memory to axiomatic inference. This section briefly outlines a new technical path as a preliminary blueprint for such a paradigm shift.

8.1 Core Idea: Anchoring Symbols to Rules

The central thesis of this paper is that symbols are merely interfaces; rules are the ontology. Current models learn only statistical associations among symbols; they never establish stable mappings between symbols and the rules behind them. Therefore, the way out lies in anchoring every symbol (or sequence of symbols) to the world rules it represents.
This anchoring requires work at two levels:

- **Concept anchoring**: binding symbols to stable concepts of things. For example, "apple" should point to an entity that has a growth cycle, is edible, and is associated with autumn. A concept is not a simple aggregation of characters; it carries the stable properties and relations that the thing has in the objective world.
- **Axiomatic constraints**: using logical rules (causality, non-contradiction, contextual matching, etc.) to constrain the relationships among concepts, so that reasoning is no longer probabilistic search but rule-based inference. The axiomatic system provides the criteria of legitimacy for the concept network, excluding logically invalid paths and guaranteeing the necessity of inference.

8.2 Architectural Blueprint: DSSCB, Axiomatic System, and Parliamentary Deliberation

Building on these ideas, the author has developed a complete neuro-symbolic architecture in

earlier work, named the “Logic Sub-Universe Parliament” [1]. Its core components include:
Dynamic Shared Symbolic Concept Bank (DSSCB): This is an evolvable concept network. Each concept node stores its definition, associated axioms, and relational edges to other concepts (e.g., implication, opposition, relevance). DSSCB is not merely a static knowledge base; it is a shared semantic space between the neural and symbolic sides—the output vectors of the neural network can be projected into the concept-vector space of DSSCB, enabling real-time interaction between probabilistic perception and symbolic reasoning.
Axiomatic System: The system is built with a set of unmodifiable root axioms, including the Axiom of Causal Consistency, the Axiom of Conceptual Consistency, the Axiom of Contextual Matching, and the Axiom of Human Well-being Primacy, among others. These axioms serve as the “constitution” of reasoning, providing hard constraints for all concept operations. Any reasoning path must pass axiomatic verification; otherwise it is rejected.
Specialized AGI Senator Network: Multiple specialized agents (such as an Ethics AGI, an Efficiency AGI, a Risk AGI) represent different perspectives and engage in debate based on the axioms and DSSCB. Each generates its own proposal, and through a conflict-driven iterative consensus protocol they refine decisions in debate, distilling robust conclusions.
Anchor-Driven Memory System (ADMS): Experiences are stored in the form of *anchors*, each containing a situational tag, core concept, axiom associations, a semantic-subgraph snapshot, and a reconstruction index. When recalling, the system uses the current axiomatic system to reconstruct the anchor into a cognition adapted to the context, achieving non-storage-based memory.
Framework Auditor AGI: An independent monitoring module that continuously supervises all reasoning processes to ensure compliance with the axiomatic system. When it encounters a fundamental conflict that the axioms cannot resolve, it proactively requests human intervention.
The key innovation of this architecture is that it no longer places the hope of “understanding” on probabilistic fitting. Instead, it installs a stable underlying framework through the axiomatic system, anchors symbols to concepts via DSSCB, and achieves multi-perspective logical verification through parliamentary debate. The neural side (large language models) is responsible only for preliminary semantic-field parsing and generating candidate concepts; the symbolic side handles concept anchoring, logical reasoning, and axiomatic verification. The two sides interact in real-time through the shared semantic space of DSSCB, enabling the leap from probability to logic.

8.3 From Probability to Logic: The “Apple in Orchard” Example Revisited

Return to the question “In which season can apples in an orchard be picked and eaten?” In the architecture proposed here, the process proceeds as follows:

- **Neural parsing**: The semantic-field recognizer (which can call an existing large language model) parses the user input into a semantic-field map *S*, containing candidate concepts: orchard, apple, pick, eat, season. The neural side completes the probabilistic collapse but only outputs candidates; it does not produce the final answer.
- **Concept anchoring**: The speaker AGI queries DSSCB according to *S*, and DSSCB returns the full definitions and relational edges for each candidate concept. For example, the node for “apple” carries: growth cycle → blossoms in spring → bears fruit in summer → ripens in autumn; edible → when ripe; compatible with “orchard,” etc.
- **Axiom-guided reasoning**: The system searches for paths in the DSSCB concept graph.

The Axiom of Causal Consistency requires that any inference be based on a verifiable causal chain; the Axiom of Contextual Matching requires that the path be compatible with the current context (orchard). Thus the path "apple → imported → available all year" is excluded, while the path "apple → growth cycle → autumn → ripe → pickable" is adopted.

- **Parliamentary deliberation and consensus**: Multiple specialized AGIs debate the reasoning result, confirming that there are no logical gaps. The Framework Auditor AGI verifies and passes the result, outputting "autumn."

In this process, each step is deterministic logical inference, not probabilistic sampling. Even if the training corpus had never contained the specific statement "apples in an orchard are pickable in autumn," the system could still derive the correct answer through axioms and the concept graph.

8.4 Why This Architecture Can Escape the Labyrinth

Compared with current mainstream approaches, the architecture proposed here has the following fundamental advantages:

Dimension	Mainstream Approaches	Proposed Architecture
Knowledge organization	Character probability distributions	Concept network (DSSCB)
Reasoning basis	Statistical correlation	Axiomatic system
Logical consistency	No guarantee	Hard axiomatic constraints
Long-text handling	Probabilistic explosion	Concept compression
Causal inference	Reduced to correlation	Causal-chain inference
Framework stability	No framework	Axioms are hardware-level constraints
Relation to symbols	Fitting co-occurrence	Anchoring to rules

The essence of this architecture is to upgrade the intelligent agent from a *probabilistic machine* to an *axiomatic engine*. Instead of trying to approximate the world's rules with infinite empirical memory, it uses finite axioms to infer infinite possibilities. This is precisely the nature of human rational intelligence and the true way out of the probabilistic labyrinth.

8.5 Engineering Path and Open Challenges

The architecture outlined here is not a castle in the air. Its core components (DSSCB, axiomatic system, parliamentary protocol, ADMS) have been described in detailed technical documents [1] and follow a modular design that can be implemented incrementally:

- The neural side can directly reuse existing large language models as semantic-field parsers; no retraining is required.
- The symbolic side can be implemented initially with rule-based engines, gradually introducing learning mechanisms to optimize the concept network and path search.

- DSSCB can start as a small-scale domain-specific concept repository and evolve through open-interface applications.
- The axiomatic system can begin with the most basic logical axioms (causality, non-contradiction, context) and later extend with domain-specific axioms.

Of course, this path still faces many challenges: how to construct an initial concept repository efficiently? How to train the projection that aligns the neural-side vectors with DSSCB's concept space? How to design an evolution mechanism for the axiomatic system? How to balance the stability of axioms with the dynamism of concepts? These questions await further exploration by the academic and engineering communities.

Nevertheless, the direction is clear: from symbol fitting to rule anchoring, from probabilistic machine to axiomatic engine. This is not a patch to existing technology; it is a reconstruction of the nature of intelligence. The analysis in this paper suggests that this is the only viable path out of the current impasse.

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9. Conclusion

In 1846, Le Verrier deduced the position of Neptune with nothing but pen and paper. He never saw the planet with his own eyes. What he relied on was not experience but rules—the mathematical expression of the law of universal gravitation. He trusted that rules were closer to the truth than direct observation.

One hundred and eighty years later, we are attempting to "see" intelligence in a different way. We have built large language models of unprecedented scale, letting them learn probabilities in an ocean of characters. They can write poetry, generate code, and hold conversations, as if they were approaching some form of "understanding." Yet when they generate "a blue sun at night," when they lose logical coherence in long texts, when they face questions that require causal reasoning, they reveal their true nature: they are merely memorizing the statistical patterns of symbols, never reaching the world rules behind those symbols.

This paper has argued from the nature of symbols that a fundamental proposition holds: symbols are merely interfaces; rules are the ontology. A ruler can measure length, but it is not length; a symbol can denote the world, but it is not the world. Current large language models have learned the graduations on the ruler without knowing what length is; they have learned the co-occurrence probabilities of symbols without knowing the rules to which the symbols refer.

This fundamental limitation has led to insurmountable difficulties for current technical approaches. Scaling merely enlarges the probability space; retrieval-augmented generation expands it with external texts; chain-of-thought generates stepwise within it; long context widens the probabilistic horizon; shallow neuro-symbolic fusion remains a "neural-first, symbolic-later" serial architecture; embodied AI substitutes physical experience for symbol probabilities. All of them are *memorizing*—the former memorizing statistical patterns of symbols, the latter memorizing causal patterns of sensation and action. No amount of memorization can leap to a rule system capable of inferring the unknown.

This paper has further argued that the only way out of this predicament is to accomplish two cognitive leaps: from symbols to concepts, and from concepts to axioms. Concepts compress the probability space, solving the global workspace problem and enabling the system to process complex information within limited capacity. The axiomatic system provides logical constraints on relationships among concepts, elevating reasoning from empirical memory to rule-based inference. Based on this approach, the paper has briefly outlined an axiomatic-driven neuro-symbolic architecture, the core of which is to use an axiomatic system as a framework and a dynamic shared symbolic concept bank as an anchor, enabling real-time interaction between the neural and symbolic sides within a unified semantic space, thereby upgrading the intelligent agent from a *probabilistic machine* to an *axiomatic engine*.

This is not a patch to existing technology; it is a reconstruction of the nature of intelligence. It is not about making AI more “human-like,” but about giving AI a cognitive architecture isomorphic to human rationality—moving from “memorizing patterns” to “understanding rules,” from “probabilistic fitting” to “rule-based inference.”

There is a detail often overlooked in Le Verrier’s story: before he calculated the position of Neptune, others had already observed the irregularities in Uranus’s orbit but could not explain them. Those observational data lay there, waiting for someone who could understand them. Today, we are faced with vast amounts of linguistic data, searching for traces of intelligence within them. But intelligence does not lie in the data; it lies in the rules behind the data. To escape the probabilistic labyrinth, what we need is the lighthouse of rules, not a brighter searchlight.

The analysis in this paper is sobering, yet clear; critical, yet constructive. It declares the end of one path while marking the beginning of another. That path is long and arduous, but it leads to genuine intelligence—the rational intelligence capable of understanding rules, inferring causality, and navigating steadily through infinite possibilities.

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